

Kartik Krishnan

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EDUCATION

Technical University of Munich (TUM) Oct 2023 — Present
B.Sc. Informatics (7th Semester) *Munich, Germany*

Gymnasium Ottobrunn 2023
Abitur (Grade: 1.7) *Ottobrunn, Germany*

PROFESSIONAL EXPERIENCE

Working Student Jun 2023 — Present
IABG - Innovation Center *Ottobrunn, Germany*

- **RC 1:10 Crawler Hardware Integration**
 - Built a Gazebo simulation of an Ackermann rover in an office environment with SLAM and Nav2, and validated it on the physical robot
 - Implemented low-level motor and steering control on an Nvidia AGX Orin (X230D carrier board) via PCA9685 PWM and an integrated ESC
 - Designed and 3D-printed a custom mount to integrate the compute hardware onto the chassis
- **Autonomous Drone Data Acquisition**
 - Developed a Python control system using the DJI Tello SDK for autonomous flight with low-latency real-time video streaming
 - Integrated OpenCV-based ArUco tag detection to navigate between stations and capture images automatically
- **Simulated Self-Parking Vehicle System**
 - Engineered an autonomous parking system in ROS 2, focusing on Ackermann-constrained path planning with Nav2
 - Implemented a lightweight alternative navigation pipeline with A* path planning, obstacle avoidance, and PID control
- **Self-Driving Model Car**
 - Trained an end-to-end driving policy via behavior cloning (PyTorch) on a Jetson Nano Jetracer (ROS Noetic), from data collection and labeling to real-time on-device inference
 - Built the full pipeline from data collection and automated labeling to model training and real-time on-device inference

PROJECTS

- **FlyBrainLab** (Hack-Nation Global AI Hackathon, 3rd Place Munich Hub): Co-developed in a team of four a multi-agent AI platform that explores the fruit fly brain using the FlyWire connectome, runs whole-brain spiking simulations (Brian2) to test circuit hypotheses, and replays results in a Three.js web viewer
- **Reframe** (TUM.ai Hackathon): Built an agent-driven pipeline that automatically reframes videos into face-aware Instagram Feed and Reel formats
- **Herdra** (THINK. MAKE. START., UnternehmerTUM): Developed in a team of five, during a two-week makeathon, an end-to-end solar-powered water-monitoring prototype for remote livestock farms, covering sensors, software, mobile app, and business case
- **iPraktikum** (with Quartett Mobile): Developed a Swift/SwiftUI navigation app that uses on-device Apple Intelligence models to suggest personalized stops based on the user's contacts
- **CPU Cache Simulation**: Implemented a CPU cache simulator in C++ with SystemC to analyze cache behavior across different configurations
- **SagiousPass** : Published a TypeScript Chrome extension for password-strength analysis and secure credential storage

SKILLS

- **Languages**: Python, C/C++, Java, Swift, TypeScript/JavaScript, HTML/CSS, RISC-V Assembly
- **Robotics**: ROS (Noetic, Humble, Jazzy), Nav2, SLAM, Gazebo (Fortress), RViz, Nvidia Jetson
- **ML & Computer Vision**: PyTorch, Torchvision, OpenCV
- **Frameworks & Tools**: Linux (Ubuntu), Git, Docker, React, Spring, SwiftUI, SystemC
- **Spoken Languages**: German (Native), English (Native), Spanish (B1)

EXTRACURRICULAR ACTIVITIES

Robotics Club, FLL Team: World Final (Detroit) and European Final (Bregenz) 2018 - 2020